

ME594 3D Robotics
HW7 Answer to selected questions

1)

(a)

$$H_{01} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(qa) & -\sin(qa) & 0 \\ 0 & \sin(qa) & \cos(qa) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{12} = \begin{bmatrix} \cos(qh) & 0 & \sin(qh) & 0 \\ 0 & 1 & 0 & w \\ -\sin(qh) & 0 & \cos(qh) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{23} = \begin{bmatrix} \cos(qk) & 0 & \sin(qk) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(qk) & 0 & \cos(qk) & -L \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b) Here are the steps

- 1) Compute homogenous transformations $H_{02} = H_{01} * H_{12}$; $H_{03} = H_{02} * H_{23}$;
- 2) Compute the position of the end-effector: $E_0 = H_{03} * [0 \ 0 \ -L \ 1]^T$; $E_0 = E_0[:3]$;
- 3) Compute the origin of frames: $o_{01} = H_{01}[0:3,3]$; $o_{02} = H_{02}[0:3,3]$; $o_{03} = H_{03}[0:3,3]$
- 4) Compute the rotation matrices: $R_{00} = \text{identity}(3 \times 3)$; $R_{01} = H_{01}[0:3,0:3]$, $R_{02} = H_{02}[0:3,0:3]$
- 5) The joint axis can be found by inspection; $n_1 = [1 \ 0 \ 0]$; $n_2 = n_3 = [0, 1, 0]$
- 6) $J_v = [\text{cross}(R_{00} * n_1, e_0 - o_{01}), \text{cross}(R_{01} * n_2, e_0 - o_{02}), \text{cross}(R_{02} * n_3, e_0 - o_{03})]$ where $\text{cross}(a, b)$ is the cross product of a and b .

$$J_v = \begin{bmatrix} 0 & -0.1264 & 0.0647 \\ 0.1173 & 0.0130 & 0.0189 \\ 0.0972 & -0.1295 & -0.1883 \end{bmatrix}$$

2)

a)

$$J = \begin{bmatrix} 0.0727 & 0.1904 & 0.1651 & -0.0047 & 0.0000 & 0.0000 \\ -0.1393 & 0.1071 & 0.0929 & -0.0006 & 0.0000 & 0.0000 \\ 0.0000 & 0.1571 & -0.0962 & 0.0022 & 0.0000 & 0.0000 \\ 0.0000 & -0.4902 & -0.4902 & 0.3881 & 0.0000 & 0.0000 \\ 0.0000 & 0.8716 & 0.8716 & 0.2183 & 0.0000 & 0.0000 \\ 1.0000 & 0.0000 & 0.0000 & 0.8954 & 0.0000 & 0.0000 \end{bmatrix}$$

b)

$$[v, \omega] = J*u$$

Gives the same answer as that obtained in HW6.

$$v = [-0.27699101 \ -0.19849493 \ -0.05908374]$$

$$\omega = [0.54118076 \ -1.60818157 \ -0.4075203]$$